CS23334 Fundamentals of Data Science Lab

Experiment 2

Exp No:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Description: Use sample data set sales-data.csv.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the data into a pandas DataFrame
file_path='C:\sales_data.csv'
df = pd.read_csv(file_path)

# Display the first few rows of the DataFrame
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

# Summary statistics
print(df.describe())

# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print(product_summary)
```

Code Continued:

```
# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()

# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
                             aggfunc=np.sum, fill_value=0)
print(pivot_table)

# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)

# Heatmap of the correlation matrix
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

```
Date       Product    Sales  Quantity Region
0 2023-01-01 Product A    200         4  North
1 2023-01-02 Product B    150         3  South
2 2023-01-03 Product A    220         5  North
3 2023-01-04 Product C    300         6  East
4 2023-01-05 Product B    180         4  West

Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64

             Sales   Quantity
count    16.000000  16.000000
mean    237.500000   5.375000
std      64.031242   1.746425
min     150.000000   3.000000
25%     187.500000   4.000000
50%     225.000000   5.500000
75%     302.500000   7.000000
max     340.000000   8.000000

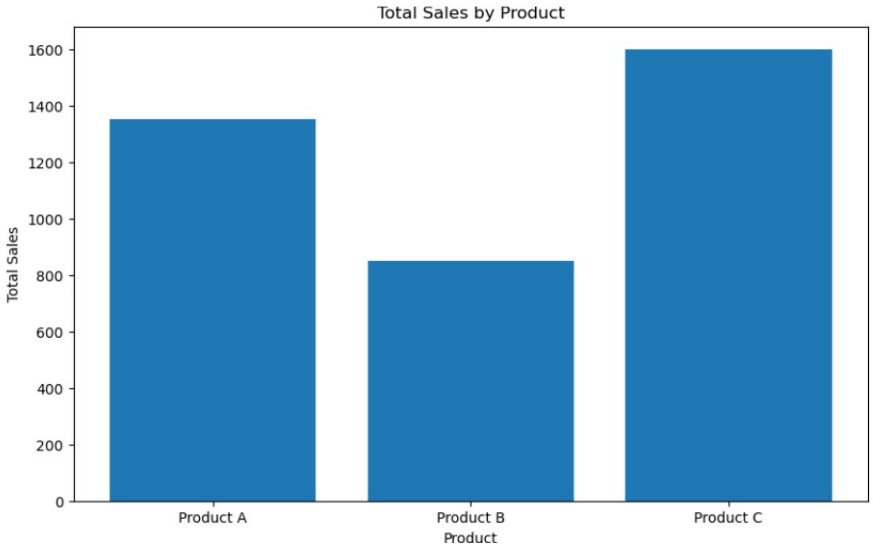
     Product   Sales  Quantity
0  Product A    1350        33
1  Product B     850        17
2  Product C    1600        36

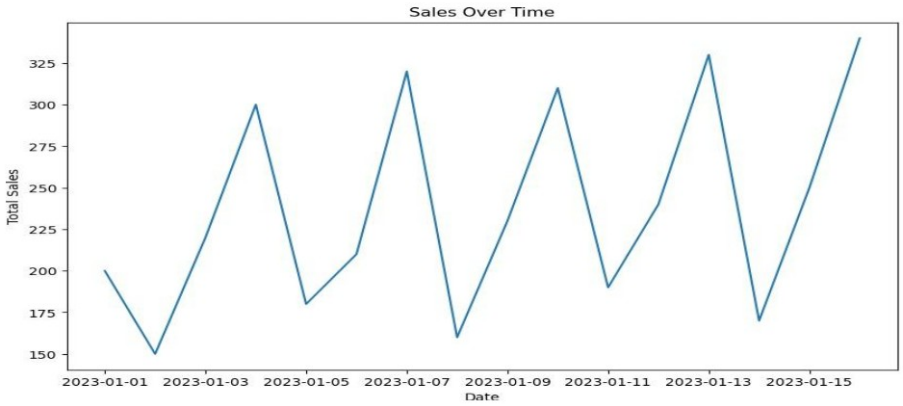
Product  Product A  Product B  Product C
Region
East             0          0       1600
North         1350          0          0
South            0        480          0
West             0        370          0

             Sales  Quantity
Sales     1.000000  0.944922
Quantity  0.944922  1.000000

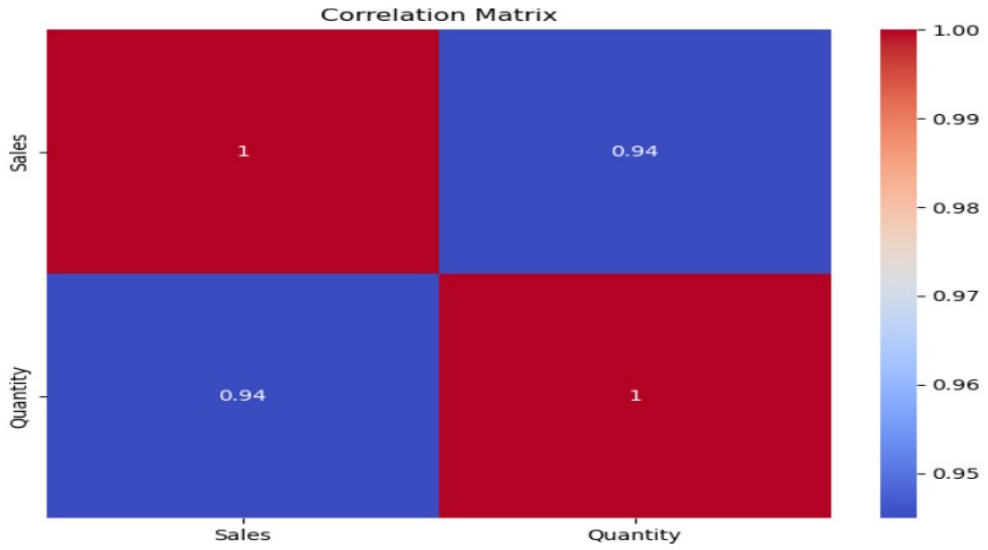
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: FutureWarning: The default
value of numeric_only in DataFrame.corr is deprecated. In a future version, it will default to
False. Select only valid columns or specify the value of numeric_only to silence this warning.
correlation_matrix = df.corr()
```

Output Figures





Output Figure Continued



Result:

The sales dataset was successfully loaded, preprocessed, and visualized. Missing values in the Sales column were imputed with the column mean, and bar and line plots revealed seasonal sales trends and top-performing product categories.